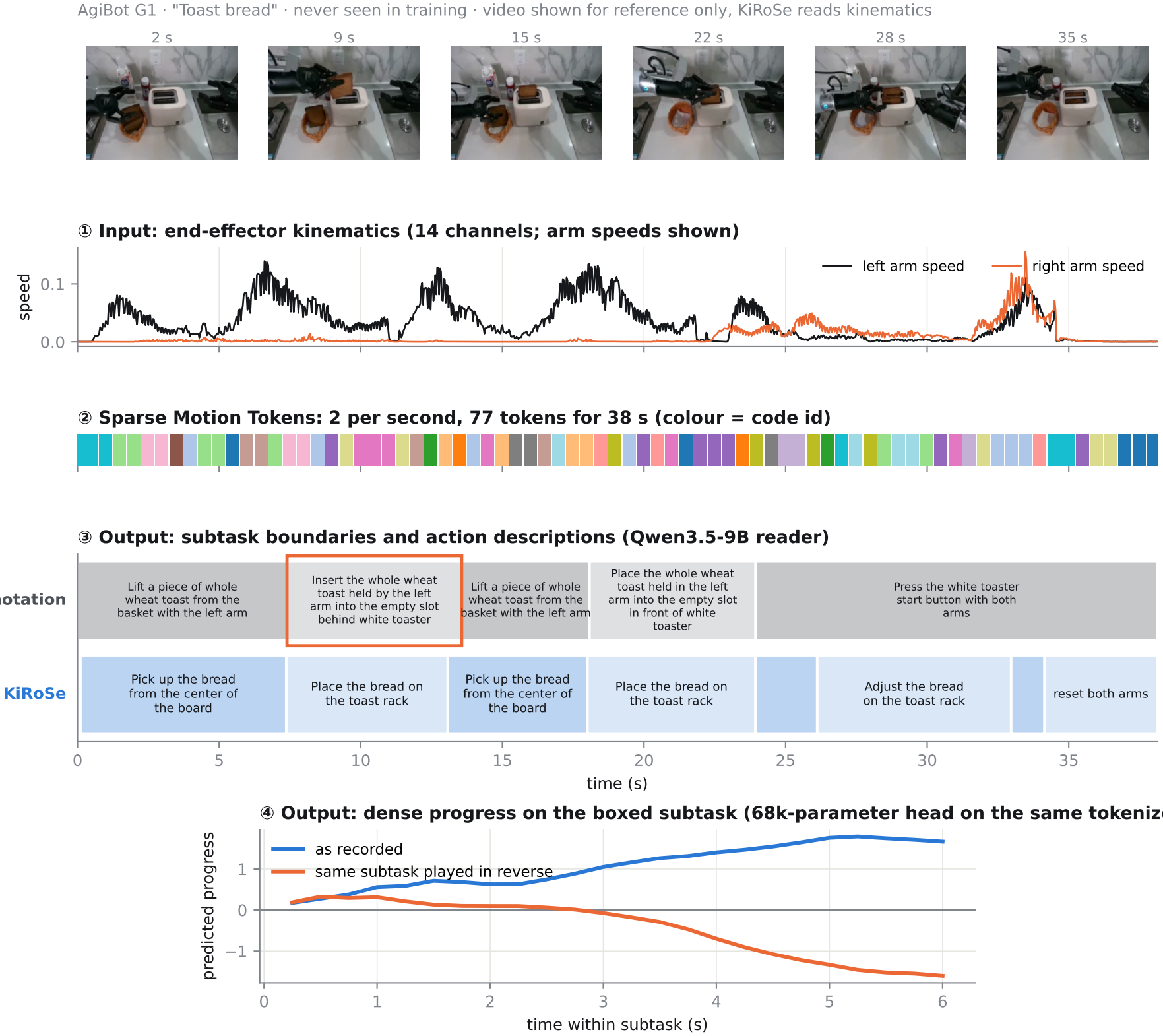

AgiBot G1 · "Toast bread" · never seen in training · video shown for reference only, KiRoSe reads kinematics
2 s
9 s
15 s
22 s
28 s
35 s
① Input: end-effector kinematics (14 channels; arm speeds shown)
speed
0.1
0.0
left arm speed
right arm speed
② Sparse Motion Tokens: 2 per second, 77 tokens for 38 s (colour = code id)
③ Output: subtask boundaries and action descriptions (Qwen3.5-9B reader)
Annotation
Lift a piece of whole wheat toast from the basket with the left arm
Insert the whole wheat toast held by the left arm into the empty slot behind white toaster
Lift a piece of whole wheat toast from the basket with the left arm
Place the whole wheat toast held in the left arm into the empty slot in front of white toaster
Press the white toaster start button with both arms
KiRoSe
Pick up the bread from the center of the board
Place the bread on the toast rack
Pick up the bread from the center of the board
Place the bread on the toast rack
Adjust the bread on the toast rack
reset both arms
0
5
10
15
20
25
30
35
time (s)
④ Output: dense progress on the boxed subtask (68k-parameter head on the same tokenizer)
predicted progress
as recorded
same subtask played in reverse
1
0
−1
0
1
2
3
4
5
6
time within subtask (s)